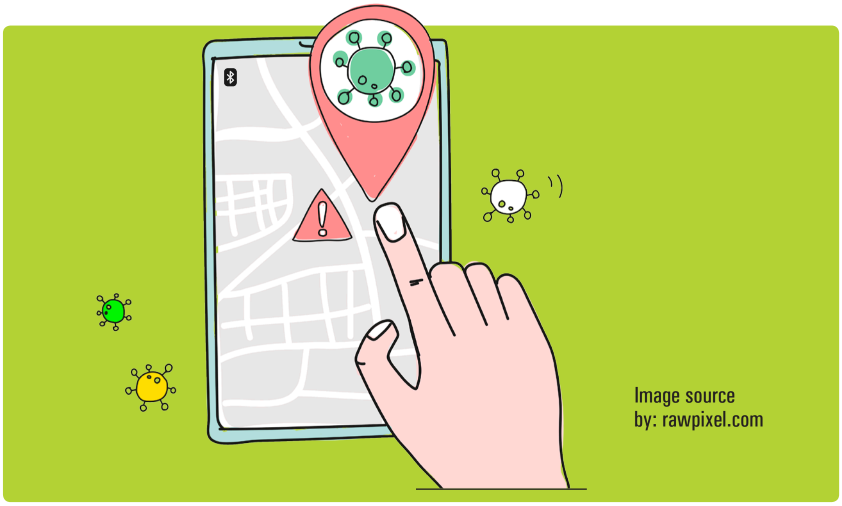
How Bluetooth Low Energy is Being Used in Covid-19 Tracing Apps

This article demystifies how Bluetooth Low Energy technology has been adapted in digital contact tracing systems to play a key role in almost all Covid-19 apps across the world.



Let us start by looking at the current context of the pandemic, understand a broad overview of Digital Contact Tracing (DCT) systems and narrow down our attention to the underlying network communication layer. We talk about how Bluetooth Low Energy (BLE) acts as the foundational technology for majority of the automated contact tracing systems although, originally, it was not envisaged for such a purpose. We also outline some of the privacy and security issues applicable for device-to-device communication using BLE.

Context

We are in midst of a pandemic that seems unrelenting. Nation states all over the world are fighting a long and strenuous battle against the devious contagion. While mass vaccination appears to be the only long-term solution, various epidemiological tools are being used by all the countries to bring some semblance of control on the unpredictable nature of its spread. One such time-tested tool is called Contact Tracing.

In the human-led contact tracing process, a representative from the health authority interviews an infected individual and tries to identify people who might have come in close contact with that person (diagnosed positive) in the last fourteen days or so (or as recommended by the epidemiologists). Subsequently, the health representative (or the contact tracer) contacts those identified individuals and informs them that they could be at the risk of being infected. They are also advised to take appropriate steps like testing, quarantining, self-isolating, etc.

Human-led contact tracing is error-prone since it depends upon the ability of the infected individual (let's call him Bob) to recollect the close contacts from past several days, many of whom could be absolute strangers to him. Moreover, when a pandemic is spreading fast, the health authority could struggle to scale up the manual contact tracing effort as it may not be possible to have so many personnel recruited and trained in a short span of time.

DCT systems have been envisaged to bridge this gap, solve the challenges of missed contacts or limited scalability, and thereby supplement the manual contact tracing process. Also, in the manual approach, the human contact tracers are expected to maintain the confidentiality of Bob's identity from his close contacts, which can never be foolproof. In a DCT system, such privacy and security measures can be implemented in a robust manner that can withstand the attacks from adversaries.

A brief overview of the architecture

Most of the current DCT systems are designed as smartphone apps that communicate with one or more back-end servers. So, in most countries in the world, Bob and his close contacts must have smartphones to take advantage of any automated contact tracing system. People who carry feature phones or do not have personal mobile devices (including many elderly individuals or children), would still have to rely upon the manual contact tracing process.